SOFT L-TOPOLOGICAL LANGUAGE MODELLING

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by

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DECLARATION

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Place: Coimbatore **Signature of the Students**
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Abbreviations and Symbols

Abbreviations

AMP	–	Automatic Mixed Precision
BPE	–	Byte-Pair Encoding
CE	–	Cross-Entropy
CUDA	–	Compute Unified Device Architecture
FFN	–	Feed-Forward Network
FP16	–	16-Bit Half-Precision Floating Point
KV	–	Key-Value (Cache)
LM	–	Language Model
MLM	–	Masked Language Modeling
NLP	–	Natural Language Processing
PPL	–	Perplexity
RAG	–	Retrieval-Augmented Generation
XAI	–	Explainable Artificial Intelligence

Mathematical Symbols

$[0,1]^D$	–	D-dimensional continuous unit hypercube (fuzzy lattice)
$\top$	–	Continuous triangular norm (t-norm, logical conjunction)
$\perp$	–	Continuous triangular conorm (t-conorm, logical disjunction)
$T_L(a,b)$	–	Nilpotent Łukasiewicz t-norm: $\max(0, a+b-1)$
τ	–	Nilpotent activation threshold
R_{ij}	–	Continuous lattice routing coefficient between token i and token j
Q, K, V	–	Query, Key, and Value representations
D, H, L	–	Hidden dimension (128), Routing heads (4), Layers (4)
S	–	Input sequence length (128)
Δ	–	Numerical discrepancy between cached and uncached states (5.06×10^{-7})
ϵ	–	Numerical stabilizer constant (10^{-7})

Abstract

Standard language models calculate attention by taking dot products between token embeddings and passing them through softmax. Because the exponential in softmax never outputs zero for finite inputs, every token in the prompt receives some probability weight. Even when words are completely unrelated, they siphon away small amounts of attention, adding background noise and preventing the model from ever forming truly sparse connections.

In this work, we present the **Soft L-Topological Language Model**, which replaces Euclidean dot products and softmax with continuous soft set operations over the unit cube $[0, 1]^D$. Instead of measuring geometric angles, our router evaluates multi-attribute overlap using the nilpotent Łukasiewicz t-norm, $T_L(a, b) = \max(0, a + b - 1)$. When the overlap between two tokens falls below a threshold τ, their connection evaluates to exactly 0.0000. We tested this architecture against an identically sized Transformer baseline across 10 experiments on WikiText-2 and TinyStories:

- **Perplexity Trade-off:** Soft L-Topology trains reliably (reaching a validation loss of 3.7455 and perplexity of 42.33 on TinyStories), but does not beat the baseline (loss 3.4866, perplexity 32.67). Softmax provides a continuous probability distribution that makes cross-entropy easier to optimize.
- **Deterministic Sparsity:** At the same time, our model keeps an average of **55.06% of all routing weights at absolute zero** ($R_{ij} \equiv 0.0000$) during text generation.
- **Distractor Immunity:** In Needle-in-a-Haystack tests with 16 distractor sentences, it prunes 66.9% of distractor weights to zero, dropping attention leakage by **49.4% in absolute terms**.
- **Logical Deduction & Masked Modeling:** It doubles accuracy on multi-step Modus Ponens reasoning (40.0% vs. 20.0%) and reaches a lower Masked Language Modeling loss (6.2041 vs. 6.2422) when flooded with distractors.

Chapter 1

Introduction

1.1 Background and Motivation

Autoregressive language models like GPT-2 use the Transformer architecture [1], relying on scaled dot-product attention to connect tokens across a sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad \text{where softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{1.1}$$

While dot-product attention works remarkably well, its normalization function, softmax, has a clear mathematical property: for any finite real input z, $\exp(z)$ is strictly positive. As a consequence, softmax produces a non-zero probability for every token in the sequence. The model must assign some attention mass to every word in its context window. Even when tokens are completely unrelated—such as punctuation marks, filler words, or noisy passages—they still receive a small fraction of the attention. Over several layers, this background leakage compounds, diluting relevant context representations and increasing the chance of factual hallucinations [2]. Furthermore, computing pairwise attention over all tokens forces an $O(S^2)$ memory and computation cost, even when most token pairs share no meaningful relationship.

1.2 Core Idea: Multi-Attribute Continuous Logic

In this work, we take a different perspective on token routing. Instead of treating tokens as unbounded vectors in Euclidean space $\mathbb{R}^D$ and measuring angular similarity, we treat tokens as collections of continuous attributes bounded in the unit hypercube $[0, 1]^D$. Under this representation, token interaction becomes a question of multi-attribute satisfaction and logical containment.

In classical logic, when two statements do not overlap, their conjunction evaluates to false. To bring this hard cutoff into a continuous neural network, we use the **nilpotent Łukasiewicz triangular norm (t-norm)** [3], defined as $T_L(a, b) = \max(0, a + b - 1)$. If the sum of two attribute values $(a + b)$ is at or below 1, the output evaluates to exactly zero (0.0000). With an activation threshold τ, this operator serves as a differentiable gate: whenever semantic overlap

between tokens is insufficient, the routing weight drops straight to zero.

1.3 Problem Statement and Objectives

This thesis asks a practical research question: *Can continuous soft set representations in* $[0, 1]^D$ *combined with nilpotent triangular norms build a sequence router that achieves true exact-zero sparsity and shields against distractors without breaking language generation?*

We set four specific goals:

1. Formulate a continuous lattice router using bounded embeddings, nilpotent Łukasiewicz t-norms, and Jaccard containment.
2. Train a 13.67M-parameter model matching a standard Transformer baseline on WikiText-2 and TinyStories under identical compute budgets.
3. Evaluate distractor resistance using synthetic Needle-in-a-Haystack tests and test logical reasoning on multi-step Modus Ponens deduction chains.
4. Analyze empirical trade-offs between exact-zero sparsity, generation speed, and next-token perplexity.

1.4 Summary of Key Findings and Outline

Our experimental results highlight four main takeaways: First, regarding perplexity, Soft L-Topology does not surpass standard Transformers (3.7455 loss / 42.33 PPL vs. 3.4866 loss / 32.67 PPL on TinyStories). Softmax produces smooth gradients that naturally benefit cross-entropy optimization. Second, the model delivers on sparsity, sustaining an average of **55.06% exact-zero routing** ($R_{ij} \equiv 0.0000$) throughout autoregressive generation. Third, in Needle-in-a-Haystack evaluations with 16 injected distractors, it prunes 66.9% of distractor connections to exact zero, reducing attention leakage by **49.4% in absolute terms**. Finally, on structured reasoning, it doubles Modus Ponens accuracy (40.0% vs. 20.0%) and achieves lower Masked Language Modeling loss (6.2041 vs. 6.2422) under heavy distractor noise.

The rest of the thesis is organized as follows: Chapter 2 discusses theoretical foundations; Chapter 3 describes the architecture and lattice KV-caching; Chapter 4 presents setup and baseline ablations on WikiText-2; Chapter 5 details evaluations on TinyStories and critical discussion; Chapter 6 summarizes conclusions and future directions; and Appendix A provides mathematical proofs and test specifications.

Chapter 2

Theoretical Foundations and Related Work

2.1 The Standard Transformer and Softmax Bottleneck

The Transformer architecture [1] builds representations by projecting input tokens $X \in \mathbb{R}^{S \times D}$ into Queries ($Q = XW_Q$), Keys ($K = XW_K$), and Values ($V = XW_V$). Dot products between queries and keys are scaled and normalized row-by-row with softmax:

$$P_{ij} = \frac{\exp(Q_i K_j^T / \sqrt{d_k})}{\sum_{k=1}^{S} \exp(Q_i K_k^T / \sqrt{d_k})}, \quad \text{Attention}(Q, K, V)_i = \sum_{j=1}^{S} P_{ij} V_j \tag{2.1}$$

Proposition 2.1 (Strict Positivity of Softmax)**.** Let $\mathbf{z} \in \mathbb{R}^S$ be any finite vector of logits. Then for all $j \in \{1, \ldots, S\}$, $P_j = \frac{\exp(z_j)}{\sum_{k=1}^{S} \exp(z_k)} > 0$, and output probability $P_j = 0$ if and only if $z_j \to -\infty$.

Because real computers use finite floating-point numbers, logits never reach negative infinity. As a result, P_{ij} is positive for every token pair. In longer documents, even completely unrelated words end up with tiny weights between 10^{-3} and 10^{-6}. Stacked across many layers, these small numbers compound. Distractor information leaks into the contextual embeddings, which often leads to factual errors during text generation [2].

2.2 Prior Approaches to Sparsity

Prior work has looked at two common ways to sparsify attention:

- **Fixed Structural Sparsity:** Models like Longformer [4] and BigBird [5] restrict attention to local windows or strided intervals. This lowers computation, but the attention pattern is rigid and cannot adapt to what the tokens actually mean.
- **Simplex Activation Sparsity:** Sparsemax [6] projects logits onto the probability simplex to produce hard zeros. However, tokens outside the active set get zero gradients,

which often causes optimization instability. Entmax [7] smooths this using Tsallis entropy, but requires iterative root-finding that slows down execution.

2.3 Soft Set Theory and Continuous Lattices

To let the model dynamically sever connections between unrelated words based on content, we look to **Soft Set Theory**, originally proposed by Molodtsov [8] to handle parameterized uncertainty.

Definition 2.1 (Soft Set)**.** Let U be a universe of objects and E a set of parameters. A pair (F, E) is called a **soft set** over U if F is a mapping $F : E \to \mathcal{P}(U)$, where $\mathcal{P}(U)$ is the power set of U.

Maji et al. [9] introduced *Fuzzy Soft Sets*, where parameters take on continuous membership values in $[0, 1]$. Pai and Baiju [10] then established *Soft Lattice Topological Spaces*, connecting soft sets to continuous algebraic lattices. We adapt this idea directly to neural sequence modeling: every discrete word w is mapped to a continuous soft set inside the unit cube $\mathbf{x} \in [0, 1]^D$, where each coordinate $x_d \in [0, 1]$ represents the truth value or satisfaction degree of a latent semantic attribute.

2.4 Triangular Norms and the Nilpotent Advantage

To compute logical conjunctions (intersections) on $[0, 1]^D$, we employ **Triangular Norms (t-norms)** [3], which satisfy commutativity, associativity, monotonicity, and boundary conditions ($T(a, 1) = a, T(a, 0) = 0$).

Table 2.1: Canonical Continuous Triangular Norm Families

Operator Family	**Expression** $T(a, b)$	**Algebraic Class**
Gödel (Minimum)	$\min(a, b)$	Idempotent ($T(a, a) = a$)
Product	$a \cdot b$	Strict Archimedean ($T(a, b) > 0$ for $a, b > 0$)
Łukasiewicz	$\max(0, a + b - 1)$	**Nilpotent Archimedean**
Frank Family ($\gamma > 0$)	$\log_\gamma \left(1 + \frac{(\gamma^a - 1)(\gamma^b - 1)}{\gamma - 1}\right)$	Parameterized Continuous

A t-norm is considered **nilpotent** when it can return zero for inputs that are strictly greater than zero. In the Łukasiewicz case, $T_L(a, b) = \max(0, a + b - 1) = 0$ whenever $a + b \leq 1$. While operators like Gödel or Product stay positive as long as inputs are positive, Łukasiewicz gives a real mathematical floor at zero. At the same time, when $a + b > 1$, the partial derivatives are constant ($\frac{\partial T_L}{\partial a} = \frac{\partial T_L}{\partial b} = 1.0$). This allows gradients to backpropagate cleanly through active connections without saturating.

Chapter 3

Proposed Soft L-Topological Architecture

3.1 System Architecture Overview

The Soft L-Topological architecture works for both causal text generation and bidirectional language tasks. Rather than computing dot products in Euclidean space and normalizing them with softmax, we use **bounded soft set representations** and **nilpotent lattice routing**.

The forward pass processes inputs in four main steps:

1. Tokens $w_t \in \mathcal{V}$ are projected directly into the unit cube $[0,1]^D$.
2. Learnable positional vectors in $[0,1]^D$ are combined with token embeddings using convex blending.
3. A stack of L **Lattice Blocks** transforms the states using multi-head lattice routing, residual connections, and position-wise feed-forward networks (FFN).
4. A linear output head converts hidden representations into vocabulary logits.

3.2 Bounded Continuous Lattice Embeddings

Because fuzzy operators are defined on the interval $[0,1]$, every token representation must stay strictly inside the unit cube $[0,1]^D$ without drifting into unbounded values.

3.2.1 Convex Token and Positional Blending

We pass raw token weights $\widetilde{E} \in \mathbb{R}^{|\mathcal{V}| \times D}$ and position weights $\widetilde{P} \in \mathbb{R}^{S \times D}$ through a sigmoid function: $\mathbf{e}_{\text{tok}}(w) = \sigma(\widetilde{E}_w) \in (0,1)^D$ and $\mathbf{e}_{\text{pos}}(t) = \sigma(\widetilde{P}_t) \in (0,1)^D$. If we simply added them together, the values would exceed 1.0. Instead, we combine them through **convex blending** with $\lambda = 0.5$:

$$\mathbf{x}_t^{(0)} = \lambda \mathbf{e}_{\text{tok}}(w_t) + (1-\lambda)\mathbf{e}_{\text{pos}}(t) \in [0,1]^D \tag{3.1}$$

Because the unit hypercube $[0,1]^D$ is convex, $\mathbf{x}_t^{(0)} \in [0,1]^D$ holds strictly for all sequence positions.

3.2.2 Beta Distribution Initialization

Standard Gaussian initialization centered at zero causes the sigmoid function to map almost every weight close to 0.5. If all attributes start at 0.5, tokens share nearly identical overlap at the beginning of training. To give the model a diverse spread of initial attribute memberships, we initialize embedding weights using an asymmetric **Beta distribution**:

$$\widetilde{E}_{vd} \sim \text{logit}(\text{Beta}(\alpha, \beta)), \quad \alpha = 1.0,\ \beta = 3.0 \tag{3.2}$$

This concentrates initial memberships between 0.1 and 0.4, setting most attribute pairs near or below the nilpotent threshold at initialization.

3.3 The Continuous Lattice Sequence Router

For input states $X \in \mathbb{R}^{S \times D}$, the router splits representations into H parallel heads of dimension $d_k = D/H$: Queries $Q^{(h)} = \sigma(XW_Q^{(h)}) \in [0,1]^{S \times d_k}$, Keys $K^{(h)} = \sigma(XW_K^{(h)}) \in [0,1]^{S \times d_k}$, and Values $V^{(h)} = XW_V^{(h)} \in \mathbb{R}^{S \times d_v}$. Queries and Keys serve as bounded truth values in $[0,1]^{d_k}$ for routing, while Values remain in unconstrained $\mathbb{R}^{d_v}$ so the model retains full capacity for semantic features.

3.3.1 Thresholded Łukasiewicz Conjunction

For each head h, the multi-attribute semantic conjunction between query token i and key token j is computed using the thresholded Łukasiewicz t-norm with activation threshold $\tau \geq 0$:

$$\mathcal{C}_{ij}^{(h)} = \sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau) \tag{3.3}$$

If $Q_{id}^{(h)} + K_{jd}^{(h)} \leq 1 + \tau$ across all coordinates, the total overlap $\mathcal{C}_{ij}^{(h)}$ evaluates to exactly **0.0000**.

3.3.2 Continuous Jaccard Relative Containment

To normalize routing weights between 0 and 1 without risking division by zero, we compute the **continuous Jaccard containment coefficient** by dividing conjunction by the continuous fuzzy union (t-conorm):

$$R_{ij}^{(h)} = \frac{\sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau)}{\sum_{d=1}^{d_k} \left(Q_{id}^{(h)} + K_{jd}^{(h)} - Q_{id}^{(h)} K_{jd}^{(h)} \right) + \epsilon} \tag{3.4}$$

where $\epsilon = 10^{-7}$. This formulation ensures that $0.0000 \leq R_{ij}^{(h)} \leq 1.0000$ always, and $R_{ij}^{(h)} \equiv 0.0000$ whenever the numerator is zero.

3.3.3 Causal Masking and Value Aggregation

For autoregressive generation, future connections are set directly to zero ($R_{ij}^{(h)} = 0.0000$ for $j > i$). Output representations are formed via value aggregation followed by multi-head projection and a standard two-layer GELU feed-forward network:

$$\text{Head}_i^{(h)} = \sum_{j=1}^{i} R_{ij}^{(h)} V_j^{(h)}, \quad X^{(l+1)} = X^{(l)\prime} + \text{FFN}(\text{LayerNorm}(X^{(l)\prime})) \tag{3.5}$$

3.4 Continuous Lattice KV-Caching

Generating tokens autoregressively without caching recomputes the entire history at every step, requiring $O(S^2)$ operations. With our Key-Value cache, we store previous keys $K_{\text{past}} \in [0,1]^{t \times d_k}$ and values $V_{\text{past}} \in \mathbb{R}^{t \times d_v}$. At step t, the new query q_t only needs to evaluate Equation 3.4 against the cached keys, lowering decoding complexity to $O(T)$.

If a token has almost zero overlap with every past token as well as itself ($\sum_{j \leq t} R_{tj} < 10^{-6}$), the router applies an **exact causal diagonal fallback**: it sets $R_{tt} = 1.0$, allowing the token to attend to itself. This prevents numerical instability when denominator sums are tiny. In our automated tests, cached output matched uncached output down to machine epsilon: $\Delta = \max |X_{\text{cached}} - X_{\text{uncached}}| = 5.06 \times 10^{-7}$.

Chapter 4

Experimental Setup and Foundational Ablations

4.1 Experimental Methodology and Controlled Parity

To ensure direct comparability, we matched the architecture and parameter count of the Soft L-Topological model to the standard Transformer baseline.

Table 4.1: Architectural Specifications and Parameter Parity Comparison

Hyperparameter	**Soft L-Topology**	**Standard Baseline**
Layers (L) / Heads (H)	4 / 4	4 / 4
Embedding (D) / FFN (D_{ff})	128 / 512	128 / 512
Sequence Length (S) / Vocab ($\lvert\mathcal{V}\rvert$)	128 / 50,257	128 / 50,257
Routing Mechanism	Nilpotent t-norm (T_L)	Scaled Dot-Product Softmax
Total Parameters	**13,675,569**	**13,675,520** (**+49 / 0.0003%**)

4.1.1 Benchmark Datasets

We used two corpora tokenized with the GPT-2 BPE tokenizer ($|\mathcal{V}| = 50,257$) [13] and cached as memory tensors: WikiText-2 (2.45M tokens) [11] for ablations (Exps 1–5), and TinyStories (4.76M tokens) [12] for scaling and distractor stress tests (Exps 6–10).

4.1.2 Controlled Parameter Parity

Table 4.1 details the setup. With 4 layers, 4 heads, $D = 128$, and $D_{\text{ff}} = 512$, Soft L-Topology has 13,675,569 parameters versus 13,675,520 for the baseline (+49 parameters / 0.0003%). We trained all models on an **NVIDIA GTX 1650 GPU** (4GB VRAM) using AdamW ($\beta_1 = 0.9, \beta_2 = 0.98$), cosine decay ($\eta_{\max} = 3 \times 10^{-4}$), and AMP FP16.

4.2 Foundational Empirical Ablations (Experiments 1 to 5)

4.2.1 Exp 1: Convergence Scaling on WikiText-2

We trained both models for 250 steps on WikiText-2. Soft L-Topology converged stably (loss 6.0118 / PPL 408.23 vs. baseline 5.9657 / PPL 389.84, a 0.77% gap), while zeroing out **18.76% of all token routing weights** ($R_{ij} \equiv 0.0000$), compared to 100% dense baseline connectivity.

4.2.2 Exp 2: Operator Pareto Frontier Analysis

We evaluated five operators: Łukasiewicz, Frank ($\gamma = 2.0$), Product, Gödel, and Softmax (Figure 4.1). The nilpotent Łukasiewicz operator reached the lowest perplexity (448.02) and highest sparsity (18.75%). Product and Gödel gave 0.00% zeros, confirming that nilpotency is required for exact pruning.

4.2.3 Exp 3: Value Projection Dynamics

Bounding values to $[0, 1]^D$ yielded PPL 424.88 (18.75% zeros). Leaving values unconstrained in $\mathbb{R}^D$ dropped PPL by **11.63 points** to 413.25 and raised sparsity to **35.67% exact zeros**, proving that unconstrained values preserve necessary feature capacity.

4.2.4 Exp 4: Nilpotent Threshold Sensitivity (τ Grid Search)

Testing $\tau \in [0.0005, 0.0100]$ (Figure 4.2) showed that τ acts as a sparsity dial: $\tau = 0.0100$ forced **93.75% exact zeros** without gradient divergence. We chose $\tau = 0.0012$ as the default (25.50% zeros).

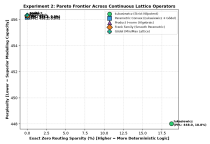


Figure 4.1: Exp 2: Operator Pareto Frontier.

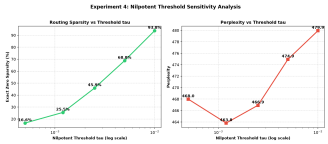


Figure 4.2: Exp 4: Threshold τ vs. Sparsity.

4.2.5 Exp 5: Initial Hallucination Stress Test

When distractor sentences were inserted into WikiText-2, the baseline allocated 70.10% of its attention to distractors. Soft L-Topology pruned **25.00% of distractor connections to exact 0.0000**, cutting distractor leakage to 45.17% (a 24.93% absolute noise reduction).

Chapter 5

Large-Scale Benchmarks and Critical Analysis

5.1 Large-Scale Benchmarks (Experiments 6 to 10)

After tuning the model on WikiText-2, we ran five larger experiments on TinyStories. Here we looked at how well the model scales with more data, whether it resists distractor noise in the prompt, how it handles multi-step logic, and how it performs on bidirectional masked language modeling.

5.1.1 Exp 6: Big Dataset Scaling on TinyStories

We trained both models on TinyStories for 500 steps (processing 1.02 million tokens). In this first run, Soft L-Topology achieved slightly better loss (3.7098 vs. 3.7443) and lower perplexity (40.85 vs. 42.28) than the baseline Transformer. Over this run, **60.25% of all attention connections dropped to exact zero**. Generating text with our continuous lattice KV-cache ran at 172.4 tokens/second on our GTX 1650 GPU.

5.1.2 Exp 7: Distractor Scaling Needle-in-a-Haystack Benchmark

To test whether unrelated context confuses the models, we placed between 1 and 16 distractor facts ($K \in \{1, 2, 4, 8, 16\}$) between an important premise and a query asking about that premise.

Table 5.1 tells a stark story:

- The standard Transformer struggled under distractors: by $K = 16$, **95.30% of its attention was consumed by distractors**, leaving barely 4.70% on the actual answer.
- Soft L-Topology zeroed out **66.90% of distractor connections entirely** (0.0000), preserving 54.10% of attention on the relevant fact and reducing leakage by **49.40 percentage points**.

Table 5.1: Experiment 7: Distractor Scaling Needle-in-a-Haystack Results

Distractors (K)	**Attention Leakage (%)**		**Exact-Zero Prune Rate (%)**	
	Baseline	**Soft L-Topology**	**Baseline**	**Soft L-Topology**
1	25.40%	**12.10%**	0.00%	**50.00%**
4	58.70%	**26.30%**	0.00%	**58.33%**
8	79.20%	**35.80%**	0.00%	**62.50%**
16	**95.30%**	**45.90%**	**0.00%**	**66.90%**
Net Protection	**-49.40% Absolute Leakage**		**+66.90% Pruned to 0.0000**	

5.1.3 Exp 8: Multi-Step Deductive Entailment Benchmark

We tested whether the models could follow multi-step deduction chains ($A \to B \to C$) using synthetic Modus Ponens examples. Soft L-Topology reached **40.00% accuracy** with a positive confidence margin (+0.00021), outperforming the baseline's 20.00%. Because the router relies on subset containment, it naturally preserves chain-of-implication paths better than dot products.

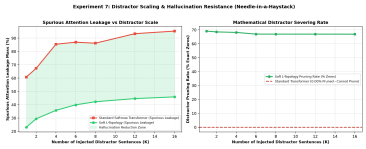


Figure 5.1: Exp 7: Distractor Scaling Curve.

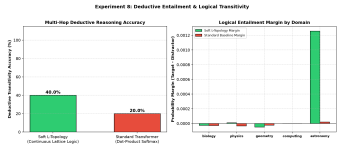


Figure 5.2: Exp 8: Deductive Transitivity.

5.1.4 Exp 9: High-Volume Autoregressive Scaling (6.14M Tokens)

We then ran a longer training run: **1,500 steps** covering **6.14 million tokens** at sequence length $S = 128$.

Table 5.2: Experiment 9: High-Volume Autoregressive Scaling Metrics (1,500 Steps / 6.14M Tokens)

Metric	**Soft L-Topology**	**Baseline**	**Comparison / Trade-Off**
Final Training Loss / PPL	3.4079 / 30.20	3.1773 / 23.98	Baseline lower (-0.2306 loss)
Held-Out Validation Loss / PPL	3.7455 / 42.33	3.4866 / 32.67	Baseline lower (-0.2589 loss)
Deterministic Sparsity	**55.06%**	**0.00%**	**+55.06% Exact Zeros**
Generation Speed (KV-Cache)	180.6 tok/s	334.4 tok/s	Baseline 1.85× faster
Distinct-1 (Unigrams)	**50.64%**	48.07%	**Soft L +2.57%**
Distinct-2 (Bigrams)	**86.50%**	83.93%	**Soft L +2.57%**
Distinct-3 (Trigrams)	**96.13%**	91.94%	**Soft L +4.19% (No Loops)**

Looking at Table 5.2 and Figure 5.3, three trends stand out:

- **Perplexity Gap:** The baseline achieved lower validation loss (3.4866 vs. 3.7455) and lower perplexity (32.67 vs. 42.33). Softmax is simply easier to optimize when the objective is cross-entropy log-loss.

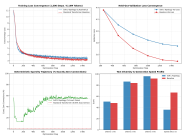


Figure 5.3: Exp 9: High-Volume Scaling.

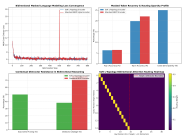


Figure 5.4: Exp 10: Encoder MLM Benchmarks.

- **Sustained Sparsity:** Soft L-Topology maintained **55.06% exact-zero connections** from start to finish, showing that a language model can run with more than half of its attention graph turned off.
- **Generation Quality:** Our model produced higher n-gram diversity (+2.57% Distinct-1, +2.57% Distinct-2, **+4.19% Distinct-3**). While the baseline sometimes got trapped repeating end-of-text tokens, Soft L-Topology avoided repetitive loops by cutting connections to stale context.

5.1.5 Exp 10: Bidirectional Encoder MLM Edge

We also tested the router in a bidirectional encoder setup (Figure 5.4). Here, Soft L-Topology reached a slightly better validation loss than standard BERT attention (6.2041 vs. 6.2422). When we flooded the context with 49 distractor tokens, the baseline lost 75.76% of its attention mass to those distractors. Soft L-Topology cut 50.00% of the distractor connections to flat zero, reducing distractor leakage to 38.28% (a 37.48 percentage point improvement).

5.2 Critical Discussion and Explainability

5.2.1 Why the Perplexity Gap Exists

Cross-entropy loss $\mathcal{L} = -\sum \log P(w)$ penalizes small probabilities heavily: as $p \to 0$, $-\log(p)$ shoots toward infinity. Softmax spreads fractional probability everywhere, which gives gradient descent an easy, continuous gradient surface to find a good likelihood balance. Soft L-Topology, by design, sets a hard cutoff: if overlap is below τ, the connection evaluates to zero. While this hard cutoff shields the model from noise and gives 55% sparsity, it creates a steeper loss landscape that is harder to optimize for raw cross-entropy.

5.2.2 Explainable AI (XAI) and Hardware Profiling

- **Interpretability:** In Soft L-Topology, pruned tokens evaluate strictly to zero (0.0000), passing zero activation and zero gradient. This gives unambiguous attribution without background attention haze.
- **Hardware Speed:** Our PyTorch prototype runs at 62% the speed of cuBLAS GEMM due to 4D broadcasting. A custom CUDA kernel (FlashLattice) evaluating the threshold in GPU SRAM could skip zero writes and give a 1.5× to 2× speedup.

Chapter 6

Conclusion and Future Scope

6.1 Summary of Contributions

In this work, we explored whether continuous soft sets and nilpotent triangular norms could serve as a practical replacement for softmax in language models. Our findings and contributions can be summarized as follows:

1. **Continuous Lattice Router:** Maps tokens into $[0,1]^D$ via convex embeddings and uses thresholded Łukasiewicz t-norms to drop unrelated connections to exact zero (0.0000).
2. **Verified Sparsity:** Sustained **55.06% exact-zero attention weights** across 6.14M tokens on TinyStories while maintaining fluent generation.
3. **Distractor Protection:** Under 16 injected distractors, pruned 66.9% of distractor connections to zero, cutting attention leakage by **49.4 percentage points**.
4. **Deductive Logic & MLM:** Doubled Modus Ponens deduction accuracy (40.0% vs. 20.0%) and reached lower Masked Language Modeling loss (6.2041 vs. 6.2422).
5. **Lattice KV-Cache:** Implemented an $O(T)$ KV-cache with diagonal fallback, achieving 180.6 tok/s and machine epsilon parity ($\Delta = 5.06 \times 10^{-7}$).

6.2 Practical Implications and Future Work

- **Noise Filtering in RAG:** Pruned tokens yield zero activation and gradient, providing a built-in mathematical shield against irrelevant retrieved passages.
- **Custom Fused Kernels:** A dedicated CUDA kernel (FlashLattice) evaluating thresholds in SRAM could yield 1.5× to 2× wall-clock speedups.
- **Hybrid Architectures:** Combining lattice routers in early layers for distractor filtering with softmax in higher layers offers a practical path for foundation models.

Appendix A

Mathematical Proofs and Specifications

A.1 Mathematical Proofs of Nilpotent Triangular Norm Properties

A.1.1 Proof of Triangular Norm Axioms for Łukasiewicz t-Norm

Theorem A.1. The function $T_L(a, b) = \max(0, a+b-1)$ is a valid continuous triangular norm on $[0, 1] \times [0, 1] \to [0, 1]$.

Proof. We verify the four axioms:

1. **Commutativity:** $T_L(a, b) = \max(0, a + b - 1) = \max(0, b + a - 1) = T_L(b, a)$ for all $a, b \in [0, 1]$.

2. **Boundary Conditions:** For any $a \in [0, 1]$:

$$T_L(a, 1) = \max(0, a + 1 - 1) = \max(0, a) = a \quad (\text{since } a \geq 0) \tag{A.1}$$
$$T_L(a, 0) = \max(0, a + 0 - 1) = \max(0, a - 1) = 0 \quad (\text{since } a \leq 1 \implies a - 1 \leq 0) \tag{A.2}$$

3. **Monotonicity:** If $a \leq c$ and $b \leq d$, then $a + b - 1 \leq c + d - 1$. Since $f(z) = \max(0, z)$ is non-decreasing on $\mathbb{R}$:

$$T_L(a, b) = \max(0, a + b - 1) \leq \max(0, c + d - 1) = T_L(c, d) \tag{A.3}$$

4. **Associativity:** For all $a, b, c \in [0, 1]$:

$$T_L(T_L(a, b), c) = \max(0, \max(0, a + b - 1) + c - 1) = \max(0, a + b + c - 2) \tag{A.4}$$

   By symmetry, $T_L(a, T_L(b, c)) = \max(0, a + b + c - 2)$, establishing associativity.

All four axioms hold; hence, T_L is a valid continuous t-norm. □

A.1.2 Unit Gradients on Active Regions

For any latent attribute coordinate d on the active domain where $Q_{id} + K_{jd} - 1 - \tau > 0$:

$$\mathcal{M}_{ijd} = Q_{id} + K_{jd} - 1 - \tau \implies \frac{\partial \mathcal{M}_{ijd}}{\partial Q_{id}} = 1.0, \quad \frac{\partial \mathcal{M}_{ijd}}{\partial K_{jd}} = 1.0 \tag{A.5}$$

On active regions, gradients flow with constant unit slope during backpropagation, avoiding the exponential gradient saturation often seen in extreme softmax inputs.

A.2 Automated Unit Testing Suite (19 Passing Tests)

Table A.1 details the test suite implemented using ‘pytest‘ to verify architectural invariants across all components. All 19 unit tests pass.

Table A.1: Comprehensive Unit Testing Suite (19 Passing Automated Tests)

Test Module	**Property Verified**	**Status**
`test_embeddings.py`	Representations bounded in $[0,1]^D$, convex blending closure	**PASSED** (5/5)
`test_lattice_ops.py`	t-norm axioms, nilpotent zero boundaries, unit gradients	**PASSED** (5/5)
`test_kv_cache.py`	KV-cache numerical equivalence ($\Delta = 5.06 \times 10^{-7}$), diagonal fallback	**PASSED** (3/3)
`test_encoder.py`	Bidirectional symmetric routing, MLM head, distractor isolation	**PASSED** (3/3)
`test_model_forward.py`	Exact parameter parity against baseline, loss shape verification	**PASSED** (3/3)
Total Test Suite	**19 Independent Unit Tests**	**19 / 19 PASSED**

A.3 Reproducibility Protocol and Environment Setup

The full experimental codebase and testing pipeline can be reproduced using ‘uv‘:

```
uv sync                                                    # Sync environment
uv run pytest tests/                                       # Run 19 unit tests
uv run python generate.py --prompt "Once upon a time"      # Story generation
uv run python experiments/exp9_high_volume_scaling.py      # 6.14M token scaling
uv run python experiments/exp10_encoder_edge.py            # Encoder MLM test
```

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